

Exploring Diabetes and Chronic Disease Rates Across U.S. States

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Background

According to the Centers for Disease Control and Prevention (2024), approximately **38 million people in the United States have diabetes**, which is frequently observed alongside other chronic diseases such as cardiovascular disease and COPD. These **co-occurring** rates differ between states.

Motivation

The purpose of this project is to conduct an exploratory data analysis (EDA) to:

- Evaluate relationships and uncover patterns between **diabetes** and other **chronic diseases** at the **state level**.
- Provide **insight** into **groups** that are disproportionately affected and the **factors** driving this disparity.

Our aim is to **inform** and **guide data scientists** onboarding to this project about how we utilize data science methods to discover and share insights that can lead to the development of more effective and efficient healthcare strategies and policies to better serve the individuals in our society.

Questions

1. What **chronic diseases** are associated with higher diabetes prevalence at the state level?
2. Which **state-level socioeconomic factors** (such as income, employment, and insurance access), and local characteristics are **most strongly associated** with **diabetes prevalence** across U.S. states?

Scope

- Population: Adults 18 and over
- Timeframe: 2022
- Location: U.S. States
- State-Level Metrics: 2022 U.S. Census Data
- Chronic Diseases Covered: Diabetes, obesity, asthma, arthritis, and chronic obstructive pulmonary disease (COPD) rates
- Granularity: One row per U.S. State

Data Sources

Primary Dataset: US Chronic Disease Indicators

The **primary dataset is the CDC's 2022 Chronic Disease Indicators**. This dataset is hosted by the Division of Population Health in the US Centers for Disease Control and Prevention (CDC) and contains indicators and metrics that are related to chronic disease.

“These indicators allow states and territories to uniformly define, collect, and report chronic disease data that are important to public health practice in their area.” (CDC, 2025)

Important Variable: a **question** (or indicator) and **measure** (such as mean, count, or proportion) for each chronic disease, **U.S. state**, and a **stratification category** to split by sex or race/ethnicity.

Location	https://catalog.data.gov/dataset/u-s-chronic-disease-indicators
Format	Comma-separated values (CSV)
Access Method	The data is available for download in different formats. We have chosen to download CSV formatted data.
Records	~84.0 MB (309,215 rows)

Secondary Dataset: Census Demographic Data- ACS

The **American Community Survey is a census that provides annual data on US people and households**. The United States Census Bureau offers an interface containing census data where users can select measures of interest and download the Data Profiles. The dataset includes only responses from the 2022 Census survey.

“Data Profiles contain the most popular social, economic, housing, and demographic data for a single geographic area.” (US Census, 2022)

Important Variables: **U.S. state** and **estimate values** of socioeconomic metrics (population, income, insurance status, employment status, and work transportation method).

Location	https://data.census.gov/table
Format	Comma-separated values (CSV)
Access Method	The data is available for download in different formats. We have chosen to download CSV formatted data.
Records	~1 MB (150,000 rows)

Data Cleaning & Manipulation

US Chronic Disease Indicators - CDC

1. Filter values to year, chronic diseases, data type, and U.S. states of interest.
2. Update the values in the 'Question' column to readable names that reflect each chronic disease.
3. Select the chronic disease columns of interest.
4. Rename the 'State' column for the future join.
5. Process each stratification (overall, sex, race/ethnicity) by **pivoting the data so each chronic disease has a single column**.
6. Merge processed data frames together.

Census Demographic Data - ACS

1. Census Data is from the Year 2022 survey.
2. Add and populate a "State" column with the appropriate state names.
3. Remove all empty columns (37) and rows (392).
4. Split the dataset into state-level and city-state-level data, **retaining only the state-level dataset for further processing**.
5. Separate the state-level data into Estimate and Margin of Error datasets and **retain only the estimate data**.
6. Removed Census locations that do not align with the scope of the project.
 - a. Guam, District of Columbia, Puerto Rico, United States, and Virgin Islands
7. Filtered the dataset to include only columns of interest (59)
8. Convert all values to numeric data types.

Merge Data

Perform a **merge** on the Final U.S. Chronic Disease Indicator and Census Demographic datasets **on "State" with an outer join**. Each row represents one U.S. state and the corresponding chronic disease and census metric



Data Cleaning & Manipulation

Data Transformations

Incomplete Data

Fields with **missing data** were identified and **removed**. The majority of these fields were estimates for **underrepresented populations** of specific **rac**es and **ethnicities** such as (but not exclusive to) Hispanic, Asian, and Pacific Islander. We made the decision to continue with the analysis without splitting by race and ethnicity and removed the corresponding columns. However, this observation uncovered **potential bias** in the data, as specific groups of individuals are systematically excluded.

Filtering and Structuring Data

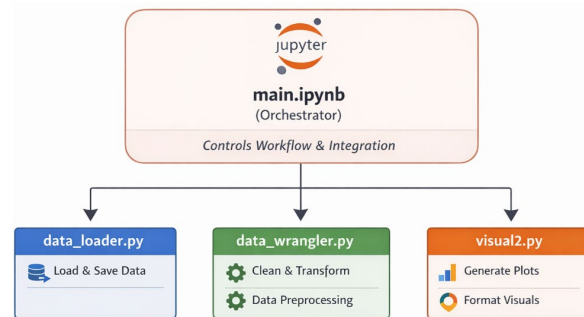
To focus the analysis on the **six subgroups** of interest for each diabetes-social factor metric, we created a column-selection that subsets the master combined DataFrame to only the relevant fields. This transformation reduces the dataset to the targeted subgroup and specific measures by filtering out nonessential columns.

Adjusting Metrics with Total Population Fields

In order to maintain a consistent population of adults only and monitor the relative percentages, population fields within the census data were leveraged to calculate total populations for certain metrics that started as percentages of a subgroup. This **standardizes** the relative metrics and allows each metric to be directly compared to diabetes prevalence across states.

Modular Structure

This approach makes the notebooks cleaner, improves code reuse, and makes the project easier to maintain and test.



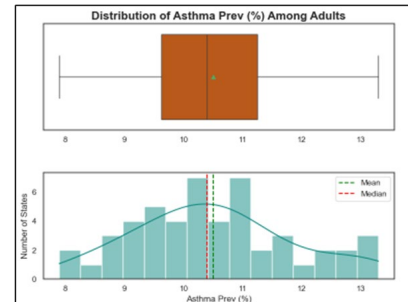
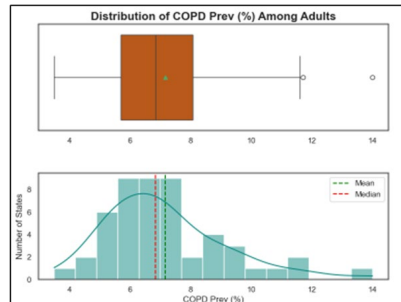
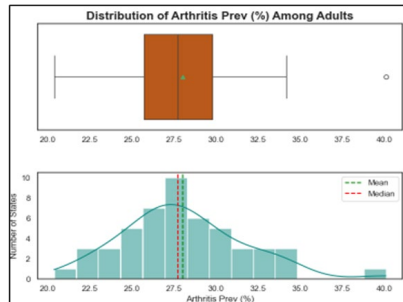
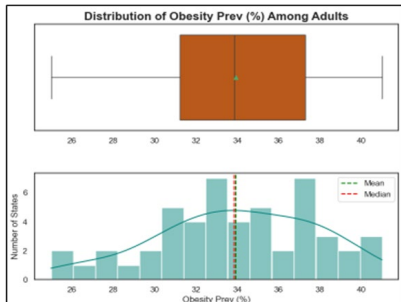
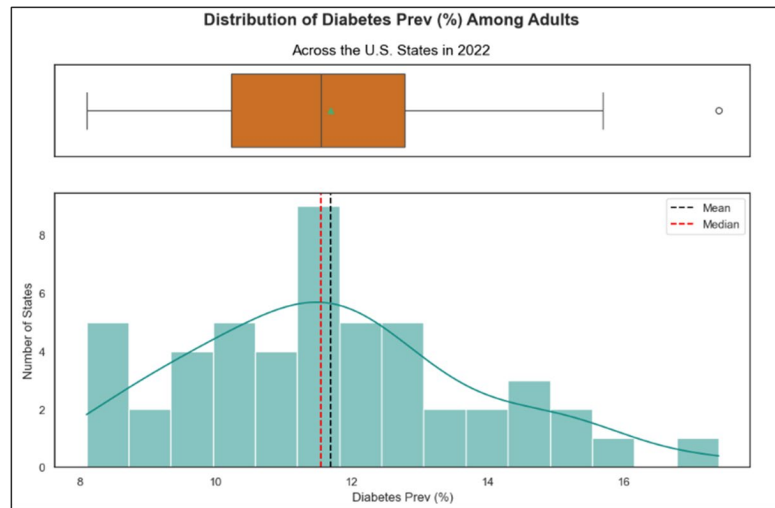
Univariate Insights - Diabetes and Other Chronic Diseases

Diabetes Prevalence

The boxplot illustrates that approximately **75%** of states have a diabetes rate **higher than 10%**. The histogram depicts the distribution is slightly **right-skewed**, suggesting that while most states cluster around similar prevalence levels, a small number of states exhibit substantially higher diabetes rates.

Outliers

Univariate analyses of **other chronic diseases** compared with diabetes prevalence showed that some distributions are **skewed** and contain **outliers**. These outliers are critical points in the data because they indicate there are states with anomalies; these particular states may be of interest in future research, so they are included in the EDA.



Diabetes and other Chronic Disease Correlations

Observations

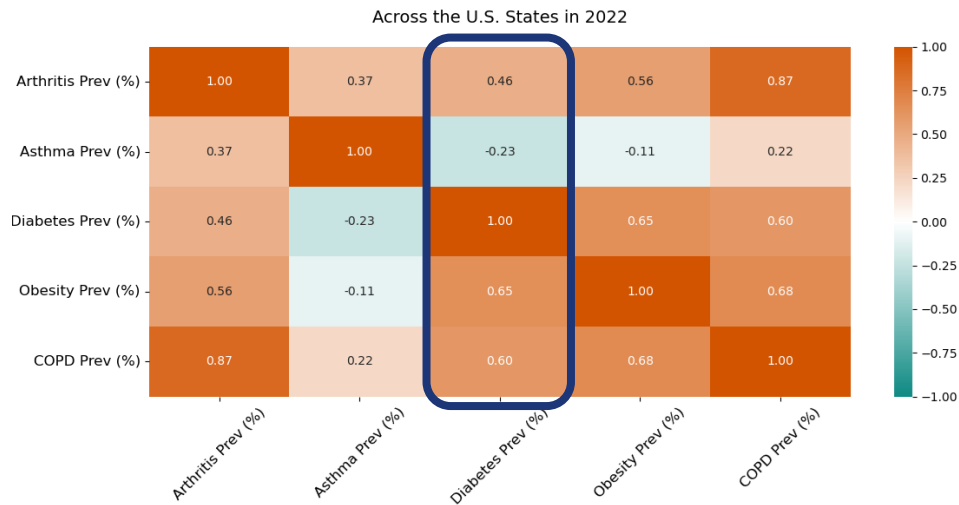
The heatmap displays the **correlations** between each **chronic disease**. The scope of this project focuses on specifically the correlations with **diabetes** prevalence (middle column). This investigates our **first research question** regarding chronic disease rates that are observed with higher diabetes prevalence.

The **Spearman Correlation Coefficient** was used to account for the monotonic relationship between diabetes and other chronic diseases. In addition, Spearman is more robust to skewed distributions and outliers, which we observed during our univariate analysis.

Obesity and **COPD** have **moderate to strong correlations** with **diabetes**, indicating states with higher obesity or COPD prevalence also tend to exhibit higher diabetes prevalence.

Asthma and **arthritis** have **weak to no correlations** with diabetes prevalence. Further analyses may show these relationships change in different subgroups; for the purpose of this project, we will continue our investigation into diabetes trends by including only obesity and COPD prevalence.

Spearman Correlation of Chronic Disease Prevalence Among Adults



Diabetes and Socioeconomic Metric Correlations

Observations

The correlations between **socioeconomic metrics** and **diabetes prevalence** are displayed in the bottom row.

The Spearman Correlation Coefficient was used again for these comparisons.

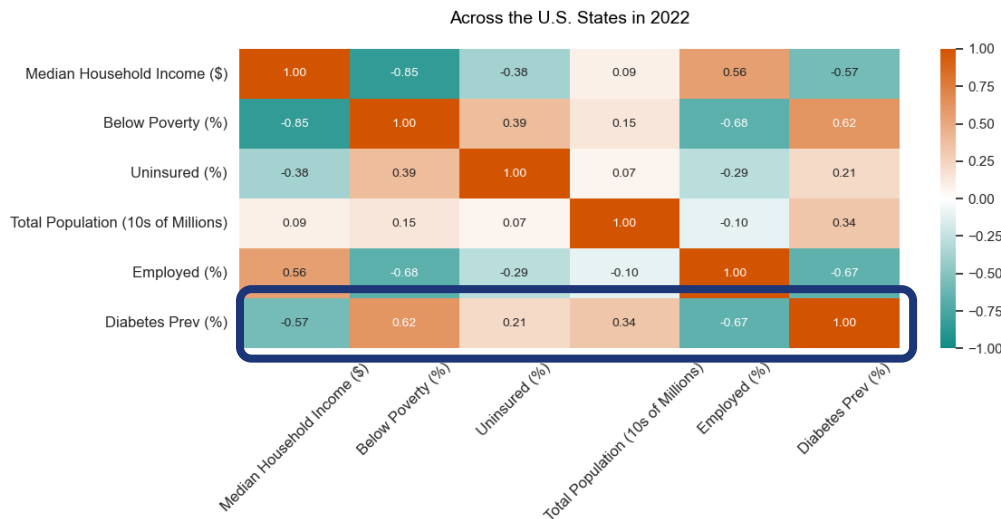
This begins to investigate **our second research question** regarding chronic disease rates and the socioeconomic metrics that are observed in tandem.

Percent of individuals **employed** has a **moderate to strong negative correlation** with diabetes prevalence. This indicates that states with a higher percentage of employed individuals tend to have a lower diabetes prevalence.

Percent of individuals **below poverty** has a **moderate to strong positive correlation** with diabetes prevalence. A state with a higher percentage of individuals below poverty also tends to have a high diabetes prevalence.

This suggests that states where the average individual makes more money *may* have better diabetes and COPD outcomes.

Spearman Correlation with Diabetes Prevalence Among Adults



Diabetes and COPD Trends Across Population

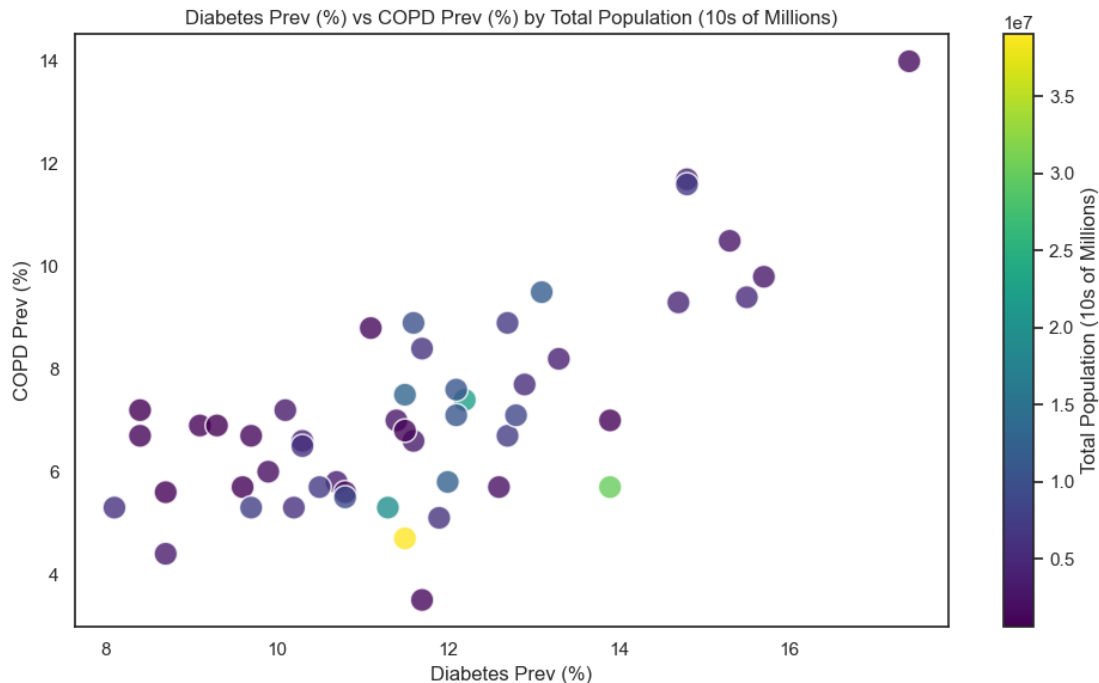
Observations

The scatterplot shows diabetes prevalence vs COPD prevalence and **colors by state population** in tens of millions.

States with **higher populations** are indicated with **lighter colors**. These observations are not clustered in a specific space, and the varying populations are scattered across the chart.

Conclusion

Overall, the scatter plot indicates that diabetes and COPD have a **consistent correlation** across states. States with bigger population have a similar diabetes and COPD prevalence trend to smaller states. This implies that that population size does not strongly change the relationship between these two conditions.



Diabetes and COPD Trends Across Employment and Poverty Rates

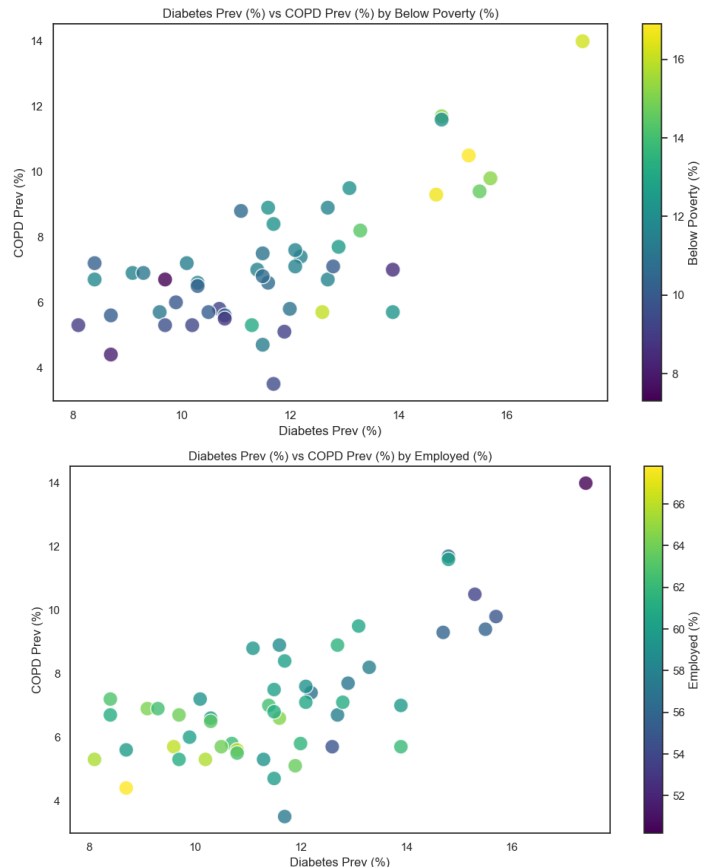
Observations

The two scatter plots show diabetes vs COPD prevalence. The top plot is colored by **percent below poverty**, and the bottom plot is colored by **percent employed**.

States with a **higher percentage of employed individuals** are clustered in the bottom left corner showing low rates of diabetes at the same time as low rates of COPD. The opposite trend is displayed for percentage below poverty; states with **lower percentage of individuals below poverty** tend to have lower prevalence of diabetes and COPD.

Conclusion

Diabetes and COPD tend to increase together across states. This pattern stays consistent when we add socioeconomic factors like employment and poverty. States with higher employment rates generally have lower levels of both diabetes and COPD, while states with higher poverty levels tend to have higher rates of both conditions. This suggests that the **socioeconomic conditions play an important role** in where diabetes and COPD are more common.



Ethical Considerations

Representative Data Bias

The chronic disease data from the CDC reports on cases of particular diagnoses, but not all cases are documented. Certain demographics may be **systematically excluded** from the data;

Low-income individuals may be less likely to see a doctor to receive a diagnosis because they cannot afford it, or households in rural areas may not have healthcare close enough.

The census data is survey-based; only individuals and households who responded to the survey are included. While the bureau makes an effort to reach as much of the population as possible, not everyone is counted and not all questions are answered.

The limitations regarding **disproportionate representation** in both of these datasets will have to be accounted for with methods such as the use of confidence intervals, splitting into subsets of more complete populations, or narrowing the scope of the project.

Risk of Resulting Decisions

When decisions are made in the interest of providing access to quality healthcare, policy makers and medical professionals need to thoroughly understand the scope of the data, who is represented, and who will be impacted positively or negatively. By performing exploratory data analysis on healthcare data, we run the **risk of these administrators making decisions about healthcare systems based on the results**, without addressing the existing bias and variance as listed above. In order to avoid the chance of discrimination, we need to be clear about what the data and resulting analyses represent, especially if it will be used for predictive modeling and additional research questions in the future.

Metric Estimations

The sources of both datasets use **models to estimate the metrics**. While some metrics include confidence intervals in the data that can be utilized for more accurate results, the modeling behind the estimates is unknown.

Protected Attributes

Some of the fields we are including in our exploratory data analysis are **protected attributes**, such as race and sex. This means they are legally guarded against discrimination, and we are required to take precautions to eliminate the risk. We can use tools such as the FairLens package in python to assess and visualize bias related to these protected classes and where it exists in our data. The results of the assessment can guide our understanding and impact regarding these attributes in our data.

Personal Health Records

Health records contain exceptionally **sensitive data**, even if they are de-identified. While our health and diagnoses data is aggregated to the state-level, it still originated from personal health records. Our data contains little to no risk of systematically identifying individuals because it is not granular, but it is still necessary to continue the conversation about the patients that the data represents, even on a high-level.

Statement of Work and References

Maria Febus

Conducted exploratory data analysis (EDA), including univariate, bivariate, and multivariate analysis. Developed visualizations, such as bar plots and histograms, to analyze distributions, and heatmaps, to identify correlations between chronic disease outcomes and state-level attributes. Performed data cleaning, handling missing values, and ensuring data quality for reliable analysis. Managed project activities by developing timelines and tracking progress against milestones.

Dung Hoang

Established the GitHub repository and designed the Python-based workflow, including a primary main.ipynb notebook supported by modular .py files for data ingestion, wrangling, and visualizations. He also developed the end-to-end data cleaning scripts for the US Census data, created the data merger script integrating the U.S. Census and Chronic Disease data, and defined the overall data processing flow.

Sara Jahanian

Identified the two datasets required for the project (U.S. Census and Chronic Disease) and applied her domain expertise to determine the appropriate data subsets needed to meet project objectives. She also developed scripts to clean and transform the U.S. Chronic Disease dataset, preparing it for a format suitable for integration with the U.S. Census data and visual creation.

References

- U.S. Census Bureau. *American Community Survey (ACS) 2022, Data Profile DP03*. Retrieved January 26, 2026, from <https://data.census.gov/table>.
- Centers for Disease Control and Prevention. 2025. *U.S. Chronic Disease Indicators (CDI)*. Retrieved January 26, 2026, from <https://catalog.data.gov/dataset/u-s-chronic-disease-indicators>.
- Centers for Disease Control and Prevention. (2024). *A report card: Diabetes in the United States*. <https://www.cdc.gov/diabetes/communication-resources/diabetes-statistics.html>